

Quarto Basics

For a demonstration of a line **plot** on a polar axis, see Figure 1.

```
import numpy as np
import matplotlib.pyplot as plt

r = np.arange(0, 2, 0.01)
theta = 4 * np.pi * r
print(theta)
print('hello')
fig, ax = plt.subplots(
    subplot_kw = {'projection': 'polar'}
)
ax.plot(theta, r)
ax.set_rticks([0.5, 1, 1.5, 2])
ax.grid(True)
plt.show()
```

```
[ 0.          0.12566371  0.25132741  0.37699112  0.50265482  0.62831853
 0.75398224  0.87964594  1.00530965  1.13097336  1.25663706  1.38230077
 1.50796447  1.63362818  1.75929189  1.88495559  2.0106193   2.136283
 2.26194671  2.38761042  2.51327412  2.63893783  2.76460154  2.89026524
 3.01592895  3.14159265  3.26725636  3.39292007  3.51858377  3.64424748
 3.76991118  3.89557489  4.0212386   4.1469023   4.27256601  4.39822972
 4.52389342  4.64955713  4.77522083  4.90088454  5.02654825  5.15221195
 5.27787566  5.40353936  5.52920307  5.65486678  5.78053048  5.90619419
 6.03185789  6.1575216   6.28318531  6.40884901  6.53451272  6.66017643
 6.78584013  6.91150384  7.03716754  7.16283125  7.28849496  7.41415866
 7.53982237  7.66548607  7.79114978  7.91681349  8.04247719  8.1681409
 8.29380461  8.41946831  8.54513202  8.67079572  8.79645943  8.92212314
 9.04778684  9.17345055  9.29911425  9.42477796  9.55044167  9.67610537
 9.80176908  9.92743279 10.05309649 10.1787602  10.3044239  10.43008761
10.55575132 10.68141502 10.80707873 10.93274243 11.05840614 11.18406985
11.30973355 11.43539726 11.56106097 11.68672467 11.81238838 11.93805208
12.06371579 12.1893795  12.3150432  12.44070691 12.56637061 12.69203432
12.81769803 12.94336173 13.06902544 13.19468915 13.32035285 13.44601656
13.57168026 13.69734397 13.82300768 13.94867138 14.07433509 14.19999879
14.3256625  14.45132621 14.57698991 14.70265362 14.82831732 14.95398103
15.07964474 15.20530844 15.33097215 15.45663586 15.58229956 15.70796327]
```

```

15.83362697 15.95929068 16.08495439 16.21061809 16.3362818 16.4619455
16.58760921 16.71327292 16.83893662 16.96460033 17.09026404 17.21592774
17.34159145 17.46725515 17.59291886 17.71858257 17.84424627 17.96990998
18.09557368 18.22123739 18.3469011 18.4725648 18.59822851 18.72389222
18.84955592 18.97521963 19.10088333 19.22654704 19.35221075 19.47787445
19.60353816 19.72920186 19.85486557 19.98052928 20.10619298 20.23185669
20.3575204 20.4831841 20.60884781 20.73451151 20.86017522 20.98583893
21.11150263 21.23716634 21.36283004 21.48849375 21.61415746 21.73982116
21.86548487 21.99114858 22.11681228 22.24247599 22.36813969 22.4938034
22.61946711 22.74513081 22.87079452 22.99645822 23.12212193 23.24778564
23.37344934 23.49911305 23.62477675 23.75044046 23.87610417 24.00176787
24.12743158 24.25309529 24.37875899 24.5044227 24.6300864 24.75575011
24.88141382 25.00707752]

```

hello

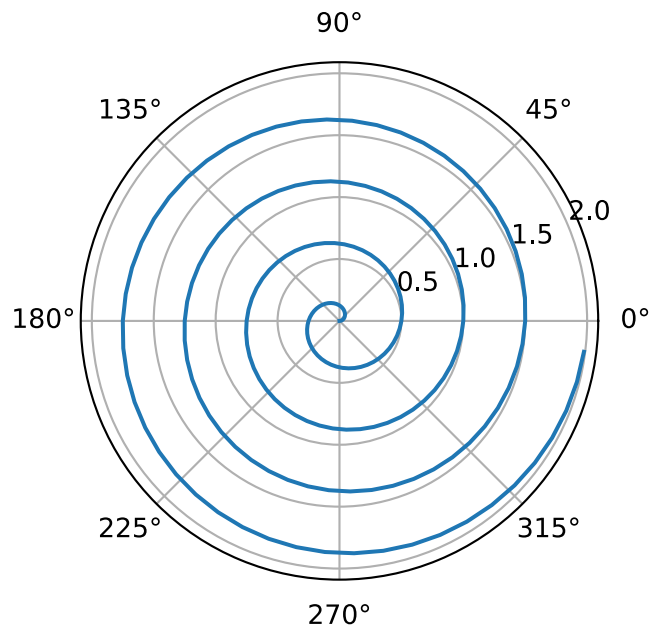


Figure 1: A line plot on a polar axis

```
print(r)
```

```

[0.  0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.1  0.11 0.12 0.13
 0.14 0.15 0.16 0.17 0.18 0.19 0.2  0.21 0.22 0.23 0.24 0.25 0.26 0.27
 0.28 0.29 0.3  0.31 0.32 0.33 0.34 0.35 0.36 0.37 0.38 0.39 0.4  0.41
 0.42 0.43 0.44 0.45 0.46 0.47 0.48 0.49 0.5  0.51 0.52 0.53 0.54 0.55
 0.56 0.57 0.58 0.59 0.6  0.61 0.62 0.63 0.64 0.65 0.66 0.67 0.68 0.69

```

0.7 0.71 0.72 0.73 0.74 0.75 0.76 0.77 0.78 0.79 0.8 0.81 0.82 0.83
0.84 0.85 0.86 0.87 0.88 0.89 0.9 0.91 0.92 0.93 0.94 0.95 0.96 0.97
0.98 0.99 1. 1.01 1.02 1.03 1.04 1.05 1.06 1.07 1.08 1.09 1.1 1.11
1.12 1.13 1.14 1.15 1.16 1.17 1.18 1.19 1.2 1.21 1.22 1.23 1.24 1.25
1.26 1.27 1.28 1.29 1.3 1.31 1.32 1.33 1.34 1.35 1.36 1.37 1.38 1.39
1.4 1.41 1.42 1.43 1.44 1.45 1.46 1.47 1.48 1.49 1.5 1.51 1.52 1.53
1.54 1.55 1.56 1.57 1.58 1.59 1.6 1.61 1.62 1.63 1.64 1.65 1.66 1.67
1.68 1.69 1.7 1.71 1.72 1.73 1.74 1.75 1.76 1.77 1.78 1.79 1.8 1.81
1.82 1.83 1.84 1.85 1.86 1.87 1.88 1.89 1.9 1.91 1.92 1.93 1.94 1.95
1.96 1.97 1.98 1.99]